

Yansen Zhang

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RESEARCH PROFILE

- Ph.D. candidate in Computer Science developing algorithmic methods for **LLM reasoning, planning, and self-evolving agents**. My research studies how LLMs can reason, plan, and improve under structured constraints and limited inference budgets, with work spanning decision reasoning, optimization modeling, scientific discovery, and data valuation. Published as first author at leading venues including *ICLR*, *ICML*, *KDD*, and *ECML PKDD*.

EDUCATION

2025.09–2025.12; **Visiting Student**, Machine Learning, Mohamed bin Zayed University of Artificial Intelligence
Supervisor: Prof. [Xue \(Steve\) Liu](#)

2022.09–Present; **Ph.D. Candidate in Computer Science**, City University of Hong Kong
Supervisor: Prof. [Chen Ma](#)

Research Areas: LLM reasoning and planning; self-evolving agents; optimization modeling; data valuation

2019.09–2022.06; **M.S. in Software Engineering**, Sun Yat-sen University
Supervisor: Prof. [Yubao Liu](#)

2015.09–2019.06; **B.S. in Software Engineering**, Northeastern University
Advisor: Prof. [Guibing Guo](#)

PROFESSIONAL EXPERIENCE

- 2026.02–Present; **Research Intern, Self-Evolving Agents**, International Digital Economy Academy (IDEA)
- Contributed to the design of **Kairos**, an end-to-end autonomous optimization agent with two stages: automatic problem modeling and algorithmic self-evolution, enabling LLMs to formulate optimization problems and iteratively refine solving strategies.
 - Designed and implemented **CostAda**, an adaptive search controller that allocates inference budgets across candidate solutions to improve search efficiency under fixed budgets.
- 2024.06–2025.08; **Research Intern, LLM Decision Reasoning**, Huawei Noah's Ark Lab
- Developed an **LLM-based explainable decision framework** (EOR) combining decision information and bipartite graph analysis to generate transparent, actionable solutions. Accepted at *ICLR 2025*.
 - Designed the **Semantic-Anchored Correction** framework (SAC-Opt) for aligning LLM-generated optimization models with problem semantics and improving modeling fidelity. Accepted at *ICML 2026*.
 - Designed **retrieval-based constraint injection methods** to incorporate external knowledge and constraints into LLM-driven forecasting and optimization pipelines. Submitted to *NeurIPS 2026*.
- 2022.04–Present; **Data Valuation and Trustworthy Recommendation**, City University of Hong Kong
- Proposed a **Shapley-value-driven data pruning framework** for estimating the contribution of training interactions, enabling principled data selection and improving recommender robustness. Accepted at *KDD 2025*.
 - Studied **diversity optimization in recommendation and retrieval systems**, including counterfactual multi-player bandits for explainable recommendation diversity and a unified taxonomy of diversification metrics and methods. Accepted at *ECML PKDD 2025*. Survey published in *TKDE 2024*.

CONFERENCE PUBLICATIONS

1. **Yansen Zhang**, Qingcan Kang, Yujie Chen, Yufei Wang, Xiongwei Han, Tao Zhong, Mingxuan Yuan, and Chen Ma (2026). *SAC-Opt: Semantic Anchors for Iterative Correction in Optimization Modeling*. In: *ICML*.
2. Ziqiang Cui, Yunpeng Weng, Xing Tang, Peiyang Liu, Shiwei Li, Bowei He, Jiamin Chen, **Yansen Zhang**, Xiquang He, and Chen Ma (2026). *Less Is More: Elevating RAG via Performance-Driven Context Compression*. In: *ICML*.
3. **Yansen Zhang**, Bowei He, Xiaokun Zhang, Haolun Wu, Zexu Sun, and Chen Ma (2025). *Counterfactual Multi-player Bandits for Explainable Recommendation Diversification*. In: *ECML PKDD*. [code].

4. **Yansen Zhang**, Qingcan Kang, Wing Yin Yu, Hailei Gong, Xiaojin Fu, Xiongwei Han, Tao Zhong, and Chen Ma (2025). *Decision Information Meets Large Language Models: The Future of Explainable Operations Research*. In: *ICLR*. [code].
5. **Yansen Zhang**, Xiaokun Zhang, Ziqiang Cui, and Chen Ma (2025). *Shapley Value-driven Data Pruning for Recommender Systems*. In: *KDD*. [code].
6. Chenhao Hu, Shuhua Huang, **Yansen Zhang**, and Yubao Liu (2022). *Learning to Infer User Implicit Preference in Conversational Recommendation*. In: *SIGIR*.
7. **Yansen Zhang**, Chenhao Hu, Genan Dai, Weiyang Kong, and Yubao Liu (2021). *Self-Adaptive Graph Neural Networks for Personalized Sequential Recommendation*. In: *ICONIP*. [code].

JOURNAL PUBLICATIONS

1. Shuliang Wang, Jiabao Zhu, Yi Wang, Chen Ma, Xin Zhao, **Yansen Zhang**, Ziqiang Yuan, and Sijie Ruan (2025). *Hierarchical Gating Network for Cross-Domain Sequential Recommendation*. *TOIS*.
2. Haolun Wu*, **Yansen Zhang***, Chen Ma, Fuyuan Lyu, Bowei He, Fernando Diaz, and Xue Liu (2024). *Result Diversification in Search and Recommendation: A Survey*. *TKDE*. [code].

ONGOING PUBLICATIONS

1. Mingyang Liu*, **Yansen Zhang***, Qingcan Kang, Xiaomin Deng, Xiongwei Han, Xiaojin Fu, Lei Chen, Tao Zhong, Mingxuan Yuan, Chen Ma, and Linqi Song (2026). "From External Information to Forecast: LLM-Based Time Series Forecasting with Constraint Modeling and Reflection Acceleration".

PATENTS

1. Yubao Liu, **Yansen Zhang** (2021). "A method for sequential recommendation model and sequential recommendation method." Chinese patent, 202111051076.3. Sun Yat-sen University.

SELECTED PROJECTS & FUNDING

HK\$100,000; **Developing an LLM-RAG Based Course Assistant to Enhance Learning Experience**, funded by the City University of Hong Kong LEAD scheme. Collaborated with a student team on retrieval-augmented generation and knowledge-graph grounding for personalized learning. Advisors: Prof. Linqi Song and Prof. Chen Ma.

TEACHING EXPERIENCE

Semester B 2025/26;	CS6487: Topics in Machine Learning , City University of Hong Kong.
Semester B 2022/23, 2023/24, 2024/25;	GE2324: The Art and Science of Data , City University of Hong Kong.
Semester A 2022/23, 2023/24, 2024/25;	CS2334: Data Structures for DS , City University of Hong Kong.
2019.10–2020.06;	Mathematical Analysis , Sun Yat-sen University.

HONOURS & AWARDS

2024–2025;	Outstanding Academic Performance Award ,	City University of Hong Kong
2021–2022;	Graduate Student Second-Class Academic Scholarship ,	Sun Yat-sen University
2019–2021;	Graduate Student First-Class Academic Scholarship ,	Sun Yat-sen University
2018;	Interdisciplinary Contest in Modeling (international-level),	Honorable Mention
2017–2018;	Model Outstanding League Member (top 5%, departmental),	Northeastern University
2015–2018;	Second Prize Scholarship ,	Northeastern University
2015–2018;	National Encouragement Scholarship ,	Northeastern University